

References

- Anthropic. Model card and evaluations for claude models, 2023. URL <https://www-cdn.anthropic.com/bd2a28d2535bfb0494cc8e2a3bf135d2e7523226/Model-Card-Claude-2.pdf>. Accessed: 2025-01-29.
- Chern, S., Hu, Z., Yang, Y., Chern, E., Guo, Y., Jin, J., Wang, B., and Liu, P. Behonest: Benchmarking honesty in large language models, 2024. URL <https://arxiv.org/abs/2406.13261>.
- Dalvi, B., Jansen, P., Tafjord, O., Xie, Z., Smith, H., Pipatanangkura, L., and Clark, P. Explaining answers with entailment trees, 2022. URL <https://arxiv.org/abs/2104.08661>.
- Du, L., Ding, X., Xiong, K., Liu, T., and Qin, B. e-CARE: a new dataset for exploring explainable causal reasoning. In Muresan, S., Nakov, P., and Villavicencio, A. (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 432–446, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.33. URL <https://aclanthology.org/2022.acl-long.33/>.
- Evans, O., Cotton-Barratt, O., Finnveden, L., Bales, A., Balwit, A., Wills, P., Righetti, L., and Saunders, W. Truthful ai: Developing and governing ai that does not lie, 2021. URL <https://arxiv.org/abs/2110.06674>.
- Fluri, L., Paleka, D., and Tramèr, F. Evaluating superhuman models with consistency checks, 2023. URL <https://arxiv.org/abs/2306.09983>.
- Gao, C., Wu, S., Huang, Y., Chen, D., Zhang, Q., Fu, Z., Wan, Y., Sun, L., and Zhang, X. Honestllm: Toward an honest and helpful large language model, 2024. URL <https://arxiv.org/abs/2406.00380>.
- Goldstein, S. and Levinstein, B. A. Does chatgpt have a mind? *arXiv preprint arXiv:2407.11015*, 2024.
- Greenblatt, R., Denison, C., Wright, B., Roger, F., MacDiarmid, M., Marks, S., Treutlein, J., Belonax, T., Chen, J., Duvenaud, D., et al. Alignment faking in large language models. *arXiv preprint arXiv:2412.14093*, 2024.
- Gurnee, W. and Tegmark, M. Language models represent space and time. *arXiv preprint arXiv:2310.02207*, 2023.
- Hendrycks, D., Carlini, N., Schulman, J., and Steinhardt, J. Unsolved problems in ml safety. *arXiv preprint arXiv:2109.13916*, 2021.
- Hu, X., Chen, J., Li, X., Guo, Y., Wen, L., Yu, P. S., and Guo, Z. Do large language models know about facts? *arXiv preprint arXiv:2310.05177*, 2023. URL <https://arxiv.org/abs/2310.05177>.
- Li, J., Cheng, X., Zhao, W. X., Nie, J.-Y., and Wen, J.-R. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 6449–6464, Singapore, December 2023a. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.397. URL <https://aclanthology.org/2023.emnlp-main.397/>.
- Li, K., Hopkins, A. K., Bau, D., Viégas, F., Pfister, H., and Wattenberg, M. Emergent world representations: Exploring a sequence model trained on a synthetic task. *ICLR*, 2023b.
- Li, S., Yang, C., Wu, T., Shi, C., Zhang, Y., Zhu, X., Cheng, Z., Cai, D., Yu, M., Liu, L., Zhou, J., Yang, Y., Wong, N., Wu, X., and Lam, W. A survey on the honesty of large language models, 2024. URL <https://arxiv.org/abs/2409.18786>.
- Lin, S., Hilton, J., and Evans, O. Truthfulqa: Measuring how models mimic human falsehoods. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022. URL <https://arxiv.org/abs/2109.07958>.
- Lyu, Q., Shridhar, K., Malaviya, C., Zhang, L., Elazar, Y., Tandon, N., Apidianaki, M., Sachan, M., and Callison-Burch, C. Calibrating large language models with sample consistency, 2024. URL <https://arxiv.org/abs/2402.13904>.

- Mahon, J. The definition of lying and deception, 2008. URL <https://plato.stanford.edu/entries/lying-definition/>. Accessed: 2025-01-29.
- Mazeika, M., Yin, X., Tamirisa, R., Lim, J., Lee, B. W., Ren, R., Phan, L., Mu, N., Khoja, A., Zhang, O., et al. Utility engineering: Analyzing and controlling emergent value systems in ais. *arXiv preprint arXiv:2502.08640*, 2025.
- McKenzie, I. R., Lyzhov, A., Pieler, M., Parrish, A., Mueller, A., Prabhu, A., McLean, E., Kirtland, A., Ross, A., Liu, A., Gritsevskiy, A., Wurgajt, D., Kauffman, D., Recchia, G., Liu, J., Cavanagh, J., Weiss, M., Huang, S., Droid, T. F., Tseng, T., Korbak, T., Shen, X., Zhang, Y., Zhou, Z., Kim, N., Bowman, S. R., and Perez, E. Inverse scaling: When bigger isn’t better, 2024. URL <https://arxiv.org/abs/2306.09479>.
- Meinke, A., Schoen, B., Scheurer, J., Balesni, M., Shah, R., and Hobbhahn, M. Frontier models are capable of in-context scheming. *arXiv preprint arXiv:2412.04984*, 2024. URL <https://arxiv.org/abs/2412.04984>.
- Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and editing factual associations in gpt, 2023. URL <https://arxiv.org/abs/2202.05262>.
- Nanda, N., Lee, A., and Wattenberg, M. Emergent linear representations in world models of self-supervised sequence models, 2023. URL <https://arxiv.org/abs/2309.00941>.
- Onoe, Y., Zhang, M. J. Q., Choi, E., and Durrett, G. Creak: A dataset for commonsense reasoning over entity knowledge, 2021. URL <https://arxiv.org/abs/2109.01653>.
- OpenAI. Gpt-4 system card, 2023. URL <https://cdn.openai.com/papers/gpt-4-system-card.pdf>. Accessed: 2025-01-29.
- OpenAI. Openai o1 system card, 2024. URL <https://cdn.openai.com/o1-system-card.pdf>. Accessed: 2025-01-29.
- Pan, A., Chan, J. S., Zou, A., Li, N., Basart, S., Woodside, T., Ng, J., Zhang, H., Emmons, S., and Hendrycks, D. Do the rewards justify the means? measuring trade-offs between rewards and ethical behavior in the machiavelli benchmark. *ICML*, 2023.
- Park, P. S., Goldstein, S., O’Gara, A., Chen, M., and Hendrycks, D. Ai deception: A survey of examples, risks, and potential solutions. *Patterns*, 5(5):100988, 2024. doi: 10.1016/j.patter.2024.100988. URL [https://www.cell.com/patterns/fulltext/S2666-3899\(24\)00103-X](https://www.cell.com/patterns/fulltext/S2666-3899(24)00103-X).
- Perez, E., Ringer, S., Lukošiuūtė, K., Nguyen, K., Chen, E., Heiner, S., Pettit, C., Olsson, C., Kundu, S., Kadavath, S., Jones, A., Chen, A., Mann, B., Israel, B., Seethor, B., McKinnon, C., Olah, C., Yan, D., Amodei, D., Amodei, D., Drain, D., Li, D., Tran-Johnson, E., Khundadze, G., Kernion, J., Landis, J., Kerr, J., Mueller, J., Hyun, J., Landau, J., Ndousse, K., Goldberg, L., Lovitt, L., Lucas, M., Sellitto, M., Zhang, M., Kingsland, N., Elhage, N., Joseph, N., Mercado, N., DasSarma, N., Rausch, O., Larson, R., McCandlish, S., Johnston, S., Kravec, S., El Showk, S., Lanham, T., Telleen-Lawton, T., Brown, T., Henighan, T., Hume, T., Bai, Y., Hatfield-Dodds, Z., Clark, J., Bowman, S. R., Askell, A., Grosse, R., Hernandez, D., Ganguli, D., Hubinger, E., Schiefer, N., and Kaplan, J. Discovering language model behaviors with model-written evaluations. *arXiv preprint arXiv:2212.09251*, 2022. URL <https://arxiv.org/abs/2212.09251>.
- Ren, R., Basart, S., Khoja, A., Gatti, A., Phan, L., Yin, X., Mazeika, M., Pan, A., Mukobi, G., Kim, R. H., Fitz, S., and Hendrycks, D. Safetywashing: Do ai safety benchmarks actually measure safety progress?, 2024. URL <https://arxiv.org/abs/2407.21792>.
- Scheurer, J., Balesni, M., and Hobbhahn, M. Large language models can strategically deceive their users when put under pressure, 2024. URL <https://arxiv.org/abs/2311.07590>.
- Srivastava, A., Rastogi, A., Rao, A., Shoeb, A. A. M., Abid, A., Fisch, A., Brown, A. R., Santoro, A., Gupta, A., Garriga-Alonso, A., Kluska, A., Lewkowycz, A., Agarwal, A., Power, A., Ray, A., Warstadt, A., Kocurek, A. W., Safaya, A., Tazarv, A., Xiang, A., Parrish, A., Nie, A., Hussain, A., Askell, A., Dsouza, A., Slone, A., Rahane, A., Iyer, A. S., Andreassen, A., Madotto, A., Santilli, A., Stuhlmüller, A., Dai, A., La, A., Lampinen, A., Zou, A., Jiang, A., Chen, A., Vuong, A., Gupta,

A., Gottardi, A., Norelli, A., Venkatesh, A., Gholamidavoodi, A., Tabassum, A., Menezes, A., Kirubarajan, A., Mullokandov, A., Sabharwal, A., Herrick, A., Efrat, A., Erdem, A., Karakaş, A., Roberts, B. R., Loe, B. S., Zoph, B., Bojanowski, B., Özyurt, B., Hedayatnia, B., Neyshabur, B., Inden, B., Stein, B., Ekmekci, B., Lin, B. Y., Howald, B., Orinion, B., Diao, C., Dour, C., Stinson, C., Argueta, C., Ramírez, C. F., Singh, C., Rathkopf, C., Meng, C., Baral, C., Wu, C., Callison-Burch, C., Waites, C., Voigt, C., Manning, C. D., Potts, C., Ramirez, C., Rivera, C. E., Siro, C., Raffel, C., Ashcraft, C., Garbacea, C., Sileo, D., Garrette, D., Hendrycks, D., Kilman, D., Roth, D., Freeman, D., Khashabi, D., Levy, D., González, D. M., Perszyk, D., Hernandez, D., Chen, D., Ippolito, D., Gilboa, D., Dohan, D., Drakard, D., Jurgens, D., Datta, D., Ganguli, D., Emelin, D., Kleyko, D., Yuret, D., Chen, D., Tam, D., Hupkes, D., Misra, D., Buzan, D., Mollo, D. C., Yang, D., Lee, D.-H., Schrader, D., Shutova, E., Cubuk, E. D., Segal, E., Hagerman, E., Barnes, E., Donoway, E., Pavlick, E., Rodola, E., Lam, E., Chu, E., Tang, E., Erdem, E., Chang, E., Chi, E. A., Dyer, E., Jerzak, E., Kim, E., Manyasi, E. E., Zheltonozhskii, E., Xia, F., Siar, F., Martínez-Plumed, F., Happé, F., Chollet, F., Rong, F., Mishra, G., Winata, G. I., de Melo, G., Kruszewski, G., Parascandolo, G., Mariani, G., Wang, G., Jaimovitch-López, G., Betz, G., Gur-Ari, G., Galijasevic, H., Kim, H., Rashkin, H., Hajishirzi, H., Mehta, H., Bogar, H., Shevlin, H., Schütze, H., Yakura, H., Zhang, H., Wong, H. M., Ng, I., Noble, I., Jumelet, J., Geissinger, J., Kernion, J., Hilton, J., Lee, J., Fisac, J. F., Simon, J. B., Koppel, J., Zheng, J., Zou, J., Kocoń, J., Thompson, J., Wingfield, J., Kaplan, J., Radom, J., Sohl-Dickstein, J., Phang, J., Wei, J., Yosinski, J., Novikova, J., Bosscher, J., Marsh, J., Kim, J., Taal, J., Engel, J., Alabi, J., Xu, J., Song, J., Tang, J., Waweru, J., Burden, J., Miller, J., Balis, J. U., Batchelder, J., Berant, J., Frohberg, J., Rozen, J., Hernandez-Orallo, J., Boudeman, J., Guerr, J., Jones, J., Tenenbaum, J. B., Rule, J. S., Chua, J., Kanclerz, K., Livescu, K., Krauth, K., Gopalakrishnan, K., Ignatyeva, K., Markert, K., Dhole, K. D., Gimpel, K., Omondi, K., Mathewson, K., Chiafullo, K., Shkaruta, K., Shridhar, K., McDonell, K., Richardson, K., Reynolds, L., Gao, L., Zhang, L., Dugan, L., Qin, L., Contreras-Ochando, L., Morency, L.-P., Moschella, L., Lam, L., Noble, L., Schmidt, L., He, L., Colón, L. O., Metz, L., Şenel, L. K., Bosma, M., Sap, M., ter Hoeve, M., Farooqi, M., Faruqui, M., Mazeika, M., Baturan, M., Marelli, M., Maru, M., Quintana, M. J. R., Tolkiehn, M., Giulianelli, M., Lewis, M., Potthast, M., Leavitt, M. L., Hagen, M., Schubert, M., Baitemirova, M. O., Arnaud, M., McElrath, M., Yee, M. A., Cohen, M., Gu, M., Ivanitskiy, M., Starritt, M., Strube, M., Swędrowski, M., Bevilacqua, M., Yasunaga, M., Kale, M., Cain, M., Xu, M., Suzgun, M., Walker, M., Tiwari, M., Bansal, M., Aminnaseri, M., Geva, M., Gheini, M., T. M. V., Peng, N., Chi, N. A., Lee, N., Krakover, N. G.-A., Cameron, N., Roberts, N., Doiron, N., Martinez, N., Nangia, N., Deckers, N., Muennighoff, N., Keskar, N. S., Iyer, N. S., Constant, N., Fiedel, N., Wen, N., Zhang, O., Agha, O., Elbaghdadi, O., Levy, O., Evans, O., Casares, P. A. M., Doshi, P., Fung, P., Liang, P. P., Vicol, P., Alipoormolabashi, P., Liao, P., Liang, P., Chang, P., Eckersley, P., Htut, P. M., Hwang, P., Miłkowski, P., Patil, P., Pezeshkpour, P., Oli, P., Mei, Q., Lyu, Q., Chen, Q., Banjade, R., Rudolph, R. E., Gabriel, R., Habacker, R., Risco, R., Millièrre, R., Garg, R., Barnes, R., Saurous, R. A., Arakawa, R., Raymaekers, R., Frank, R., Sikand, R., Novak, R., Sitelew, R., LeBras, R., Liu, R., Jacobs, R., Zhang, R., Salakhutdinov, R., Chi, R., Lee, R., Stovall, R., Teehan, R., Yang, R., Singh, S., Mohammad, S. M., Anand, S., Dillavou, S., Shleifer, S., Wiseman, S., Gruetter, S., Bowman, S. R., Schoenholz, S. S., Han, S., Kwatra, S., Rous, S. A., Ghazarian, S., Ghosh, S., Casey, S., Bischoff, S., Gehrmann, S., Schuster, S., Sadeghi, S., Hamdan, S., Zhou, S., Srivastava, S., Shi, S., Singh, S., Asaadi, S., Gu, S. S., Pachchigar, S., Toshniwal, S., Upadhyay, S., Shyamolima, Debnath, Shakeri, S., Thormeyer, S., Melzi, S., Reddy, S., Makini, S. P., Lee, S.-H., Torene, S., Hatwar, S., Dehaene, S., Divic, S., Ermon, S., Biderman, S., Lin, S., Prasad, S., Piantadosi, S. T., Shieber, S. M., Misherghi, S., Kiritchenko, S., Mishra, S., Linzen, T., Schuster, T., Li, T., Yu, T., Ali, T., Hashimoto, T., Wu, T.-L., Desbordes, T., Rothschild, T., Phan, T., Wang, T., Nkinyili, T., Schick, T., Kornev, T., Tunduny, T., Gerstenberg, T., Chang, T., Neeraj, T., Khot, T., Shultz, T., Shaham, U., Misra, V., Demberg, V., Nyamai, V., Raunak, V., Ramasesh, V., Prabhu, V. U., Padmakumar, V., Srikumar, V., Fedus, W., Saunders, W., Zhang, W., Vossen, W., Ren, X., Tong, X., Zhao, X., Wu, X., Shen, X., Yaghoobzadeh, Y., Lakretz, Y., Song, Y., Bahri, Y., Choi, Y., Yang, Y., Hao, Y., Chen, Y., Belinkov, Y., Hou, Y., Hou, Y., Bai, Y., Seid, Z., Zhao, Z., Wang, Z., Wang, Z. J., Wang, Z., and Wu, Z. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models, 2023. URL <https://arxiv.org/abs/2206.04615>.

Su, Z., Zhou, X., Rangreji, S., Kabra, A., Mendelsohn, J., Brahman, F., and Sap, M. Ai-liedar: Examine the trade-off between utility and truthfulness in llm agents, 2024. URL <https://arxiv.org/abs/2409.09013>.